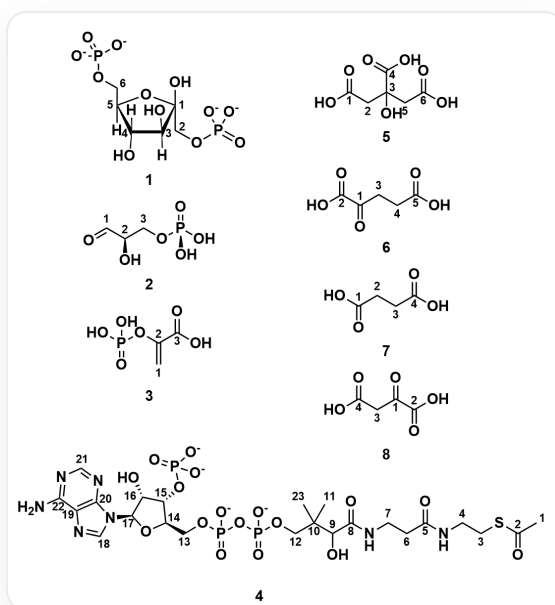


Question

There is now a sample of [1- ^{14}C] glucose, in which C – 1 is labeled with the ^{14}C isotope. Bacterial culture medium is added to it, and after a period of time, the following metabolic intermediates $i = 1 \sim 8$ are separated (with atom numbering):



Numbering of intermediates and their corresponding SMILES: 1. O[C@@:1]1([C:2]P([O-])([O-])=O)[C@@:3]([H])(O)[C@@:4](O)[C@:5]([C:6]OP([O-])([O-])=O)([H])O1 2. O=[C:1][C@H:2](O)[C:3]OP(O)(O)=O 3. [C:1]=[C:2](OP(O)(O)=O)[C:3](O)=O 4. [C:1][C:2](S[C:3][C:4]N[C:5]([C:6][C:7]N[C:8]([C:9]([C:10]([C:11])([C:12]OP(OP(O[C:13][C@@H:14]1[C@@H:15](OP([O-])([O-])=O)[C@@H:16](O)[C@H:17](N2[C:18]=N[C:19]3=[C:20]2N=[C:21]N=[C:22]3N)O1)([O-])=O)([O-])=O)[C:23])O)=O)=O)=O 5. O=[C:1](O)[C:2][C:3]([C:4](O)=O)(O)[C:5][C:6](O)=O 6. O=[C:1]([C:2](O)=O)[C:3][C:4][C:5](O)=O 7. O=[C:1](O)[C:2][C:3][C:4](O)=O 8. O=[C:1]([C:2](O)=O)[C:3][C:4](O)=O

For each metabolic intermediate i , based on your understanding of the glucose metabolic pathway, find the atom numbers of the sites where the ^{14}C label may appear, and define a set P_i , which represents the set composed of the atom numbers of all carbon atoms that may be labeled with ^{14}C in the i -th metabolic intermediate.

* For example: If in product i , the label may appear on carbon atoms 2 and 3, then $P_i = \{2, 3\}$.

The calculation rule for z_i is divided into two cases according to the size of the set P_i (i.e., the number of possible labeling sites):

$$z_i = \begin{cases} \frac{p}{i^2 + 1} & \text{if there is only one possible labeling site (i.e. } |P_i| = 1, \text{ and } P_i = \{p\}) \\ \frac{\prod_{p \in P_i} p}{i^2} & \text{if there are multiple possible labeling sites (i.e. } |P_i| > 1) \end{cases}$$

* In the second case, $\prod_{p \in P_i} p$ represents the product of all atom numbers in the set P_i .

Finally, calculate the sum S of all z_i values:

$$S = \sum_{i=1}^8 z_i$$

Choose the correct option from the following options, keeping the final result to four decimal places, and choose the option whose deviation from your calculation result is within 1%, otherwise choose option **A**: All other options are incorrect.

A. All other options are incorrect

B. 1.836

C. 2.100

D. 2.210

E. 2.276

F. 2.311

G. 2.353

H. 2.418

I. 2.436

J. 2.670

K. 3.604

Answer

Correct Answer: **I**

Detailed Explanation

The core of this question is to track the fate of the ^{14}C label from $[1-^{14}\text{C}]$ glucose in the major metabolic pathways (glycolysis in the cytoplasm, tricarboxylic acid cycle in the mitochondrial matrix). First, it is necessary to identify the names of the 8 intermediates in the picture and understand their reactions in the metabolic pathways:

1. Fructose-1,6-bisphosphate: This is a key intermediate in the glycolysis pathway. After consuming two molecules of ATP, glucose is converted into this highly activated 6-carbon sugar molecule. Its formation is an important and irreversible regulatory point in the glycolysis pathway.

CHECKPOINT

0.5 PTS

Intermediate 1 is Fructose-1,6-bisphosphate

2. Glyceraldehyde-3-phosphate (G3P): Fructose-1,6-bisphosphate is cleaved by aldolase into two 3-carbon molecules, one of which is glyceraldehyde-3-phosphate. It is the direct substrate for subsequent reactions in glycolysis, marking the beginning of the 3-carbon stage.

CHECKPOINT

0.5 PTS

Intermediate 2 is Glyceraldehyde-3-phosphate (G3P)

3. Phosphoenolpyruvate (PEP): A high-energy intermediate at the end of the glycolysis pathway. It generates one molecule of ATP by transferring a high-energy phosphate group to ADP (substrate-level phosphorylation), and itself

is converted to pyruvate, which is the final product of glycolysis.

CHECKPOINT

0.5 PTS

Intermediate 3 is Phosphoenolpyruvate (PEP)

4. Acetyl-CoA: Pyruvate produced by glycolysis enters the mitochondria and is catalyzed by the pyruvate dehydrogenase complex in the mitochondrial matrix, losing one carbon atom (released as CO_2) and binding to coenzyme A to form 2-carbon acetyl-CoA. This is the key bridge molecule connecting glycolysis and the tricarboxylic acid cycle.

CHECKPOINT

0.5 PTS

Intermediate 4 is Acetyl-CoA

5. Citrate: The first step of the tricarboxylic acid cycle. Acetyl-CoA (2 carbons) transfers its acetyl group to oxaloacetate (4 carbons), condensing to form a 6-carbon citrate molecule, thus starting the cycle.

CHECKPOINT

0.5 PTS

Intermediate 5 is Citrate

6. α -Ketoglutarate: Citrate is converted into this 5-carbon intermediate through a series of reactions (including two decarboxylations, i.e., releasing two molecules of CO_2). It is not only part of the cycle, but also an important precursor for the synthesis of amino acids such as glutamate.

CHECKPOINT

0.5 PTS

Intermediate 6 is α -Ketoglutarate

7. Succinate: α -Ketoglutarate is decarboxylated again to produce 4-carbon succinate. This step is accompanied by the production of one molecule of GTP (or ATP), which is another substrate-level phosphorylation in the tricarboxylic acid cycle.

CHECKPOINT

0.5 PTS

Intermediate 7 is Succinate

8. Oxaloacetate (OAA): Succinate is regenerated into 4-carbon oxaloacetate through several oxidation reactions. Oxaloacetate, as the end point of the cycle, is also the starting point of the next cycle. It will combine with new acetyl-CoA again to start a new round of the cycle.

CHECKPOINT

0.5 PTS

Intermediate 8 is Oxaloacetate (OAA)

These intermediates constitute a continuous chain of biochemical reactions: Glucose $\rightarrow$ [Glycolysis] $\rightarrow$ Fructose-1,6-bisphosphate $\rightarrow$ Glyceraldehyde-3-phosphate $\rightarrow$ Phosphoenolpyruvate $\rightarrow$ Pyruvate $\rightarrow$ [Link Reaction] $\rightarrow$ Acetyl-CoA + Oxaloacetate $\rightarrow$ [Tricarboxylic Acid Cycle] $\rightarrow$ Citrate $\rightarrow$ α -Ketoglutarate $\rightarrow$ Succinate $\rightarrow$... $\rightarrow$ (Regenerated) Oxaloacetate.

Based on the atomic labeling in the picture, combined with the metabolic pathway of glucose, the carbon atom number that may contain isotopic labeling in each intermediate can be inferred:

1. Fructose-1,6-bisphosphate: The label is located on the carbon at the 1 position of fructose in 1,6-bisphosphate, which is carbon atom number 2 in the figure.

$$z_1 = \frac{p}{i^2 + 1} = \frac{2}{1^2 + 1} = \frac{2}{2} = 1$$

CHECKPOINT

1 PTS

The label of Fructose-1,6-bisphosphate is located on carbon atom number 2, $z_1 = 1$

2. Glyceraldehyde-3-phosphate: The label is located on carbon atom number 3, which is linked to phosphate.

$$z_2 = \frac{p}{i^2 + 1} = \frac{3}{2^2 + 1} = \frac{3}{5} = 0.6$$

CHECKPOINT

1 PTS

The label of Glyceraldehyde-3-phosphate is located on carbon atom number 3, $z_2 = 0.6$

3. Phosphoenolpyruvate (PEP): The label is located at the β position of the unsaturated ketone, which is carbon atom number 1 in the figure.

$$z_3 = \frac{p}{i^2 + 1} = \frac{1}{3^2 + 1} = \frac{1}{10} = 0.1$$

CHECKPOINT

1 PTS

The label of Phosphoenolpyruvate (PEP) is located on carbon atom number 1, $z_3 = 0.1$

4. Acetyl-CoA: The label is located on the methyl group of the acetyl group, which is carbon atom number 1 in the figure.

$$z_4 = \frac{p}{i^2 + 1} = \frac{1}{4^2 + 1} = \frac{1}{17}$$

CHECKPOINT

1 PTS

The label of Acetyl-CoA is located on carbon atom number 1, $z_4 = \frac{1}{17}$

5. Citrate: The label is located on the carbon α to the carboxyl group with two methylene groups, which are carbon atoms number 2 and 5 in the figure.

$$z_5 = \frac{\prod_{p \in P_5} p}{i^2} = \frac{2 \times 5}{5^2} = \frac{10}{25} = 0.4$$

CHECKPOINT

1 PTS

The label of Citrate is located on carbon atoms number 2 and 5, $z_5 = 0.4$

6. α -Ketoglutarate: The label is located on the carbon α to the carboxyl group with a methylene group, which is carbon atom number 4 in the figure.

$$z_6 = \frac{p}{i^2 + 1} = \frac{4}{6^2 + 1} = \frac{4}{37}$$

CHECKPOINT

1 PTS

The label of α -Ketoglutarate is located on carbon atom number 4, $z_6 = \frac{4}{37}$

7. Succinate: The label is located on the carbon α to the carboxyl group with two methylene groups, which are carbon atoms number 2 and 3 in the figure.

$$z_7 = \frac{\prod_{p \in P_7} p}{i^2} = \frac{2 \times 3}{7^2} = \frac{6}{49}$$

CHECKPOINT

1 PTS

The label of Succinate is located on carbon atoms number 2 and 3, $z_7 = \frac{6}{49}$

8. Oxaloacetate: The label is located on the two middle carbons of the carbon chain, which are carbon atoms number 1 and 3 in the figure.

$$z_8 = \frac{\prod_{p \in P_8} p}{i^2} = \frac{1 \times 3}{8^2} = \frac{3}{64}$$

CHECKPOINT

1 PTS

The label of Oxaloacetate is located on carbon atoms number 1 and 3, $z_8 = \frac{3}{64}$

Finally, we add up all the calculated z_i values to get the sum S .

$$S = \sum_{i=1}^8 z_i = z_1 + z_2 + z_3 + z_4 + z_5 + z_6 + z_7 + z_8$$

$$S = 1 + 0.6 + 0.1 + \frac{1}{17} + 0.4 + \frac{4}{37} + \frac{6}{49} + \frac{3}{64}$$

$$S \approx 1 + 0.6 + 0.1 + 0.0588 + 0.4 + 0.1081 + 0.1224 + 0.0469$$

$$S \approx 2.4362$$

Therefore, the sum S of z_i of all intermediates is approximately 2.4362.

CHECKPOINT

1 PTS

The sum of z_i of all intermediates $S \approx 2.4362$